

Chloe Barker

214-733-7514 | chloedbarkerr@icloud.com | Dallas, TX
Linkedin: [linkedin.com/in/chloedbarker](https://www.linkedin.com/in/chloedbarker) | Portfolio: chloedbarker.github.io

SUMMARY

Data scientist completing an M.S. in Data Science (Aug 2026) with experience applying machine learning and statistical analysis to business and research questions. Uses Python, Databricks, and Snowflake across insurance, healthcare, and environmental data, with strengths in interpretable modeling and stakeholder communication.

EDUCATION

Southern Methodist University | Dallas, TX Jan 2025 - Present
M.S. in Data Science | GPA: 3.97 | Expected Aug 2026
The University of Texas at Austin | Austin, TX Aug 2020 - May 2024
B.S.A. in Biology | Minor: Health Communication | Pre-Health Professions Certificate | GPA: 3.77

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, NumPy, scikit-learn), SQL, R (tidyverse, caret), SAS
Machine Learning & Statistics: Regression, classification, gradient boosting, clustering, time series forecasting, NLP, feature engineering, cross-validation, hyperparameter tuning, model evaluation, hypothesis testing, A/B testing
Data Platforms & Management: Databricks, Snowflake, PySpark, MySQL, data pipelines, data cleaning and transformation, relational database design, Git/GitHub
Visualization & AI: Tableau, matplotlib, seaborn, Plotly, R Shiny, LLM applications, RAG

EXPERIENCE

Goosehead Insurance | Westlake, TX May 2026 - Present
Data Science Intern

- Developed and evaluated machine-learning models using insurance data in Snowflake and Databricks to support lead prioritization and marketing analytics.
- Prepared and transformed large-scale datasets using SQL, Python, and PySpark; compared classification approaches and communicated findings to business stakeholders.

Applied Analytics Group, LLC | Remote Jan 2026 - May 2026
Data Science Analyst (Volunteer)

- Supported early-stage analytics projects using Python, R, and SQL, spanning data cleaning, exploratory analysis, and reporting.
- Standardized and documented technical workflows to improve reproducibility, clarity, and handoff across the team.

Outlier AI | Remote Aug 2024 - Jan 2026
AI Math Evaluator

- Assessed AI-generated mathematical solutions across algebra, calculus, and statistics, documenting reasoning errors and grading outputs for correctness, clarity, and completeness.
- Created structured test cases and error analyses to strengthen the reliability of reasoning-based AI systems.

RUSH University, Dept. of Microbial Pathogens & Immunity | Chicago, IL Summers 2021 & 2023
Research Assistant

- Investigated alcohol, circadian, and HIV-medication effects on the gut microbiome and bone density using longitudinal clinical data in R and Excel.
- Prepared, visualized, and documented experimental data through reproducible workflows for multidisciplinary biomedical research teams.

SELECTED PROJECTS

M.S. Capstone: Groundwater Lithium & Dementia Prevalence | Python

- Modeled dementia prevalence across 254 Texas counties using groundwater and dementia data, engineering multi-year exposure measures and validating results through robustness and sensitivity checks.
- Visualized county-level geographic patterns and presented findings to faculty; received the MSDS Best Capstone Presentation award.

LLM/RAG Document Q&A Application | Databricks, Python

- Created a retrieval-augmented application that answers questions about PDF documents by retrieving relevant passages and generating source-grounded responses.
- Implemented an interactive interface with document processing, retrieval, prompting, and secure API-key management in Databricks.

Diabetes Readmission Triage | Python, scikit-learn

- Developed a risk-ranking pipeline across 71,518 hospital encounters to prioritize patients for limited transitional-care capacity while preventing data leakage.
- Benchmarked models with capacity-aware metrics and translated results into staffing scenarios and stakeholder recommendations.